

Feeding a Nation

Regional Food Inflation, Welfare, and Trade Regimes
under Indonesia's Free Nutritious Meals Program

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Universitas Padjadjaran · Dewan Ekonomi Nasional · Bank Indonesia
EcoMod2026 · University of Luxembourg, Luxembourg · 8–10 July 2026

Motivation

MBG at a Glance: A Fiscal Event, Not Just a Meal Program

USD 20B

2026 budget

9%

of state expenditure

83M

recipients

34

provinces

- ▶ Indonesia's *Makan Bergizi Gratis* (MBG): launched January 2025, flagship program of the Prabowo administration — one of the largest single fiscal commitments in the national budget.
- ▶ Targets pregnant women, toddlers, and school children from primary through upper-secondary school.

For context: total global school-feeding expenditure by domestic governments $\approx$ USD 48bn (2022, all countries combined). India's Midday Meal Scheme reaches $\approx$ 100M but covers only primary/lower-secondary. Brazil $\approx$ 40M since the 1950s. South Korea $\approx$ 7M. MBG's combination of universal provision, wide age coverage, and fiscal scale is without precedent globally.

The Policy Tension: Feeding 83 Million amid a Legacy of Trade Protection

- Five decades of Indonesian trade policy: persistent oscillation between liberalisation and protection.
- Protectionist tendencies reassert themselves under economic shocks, political transitions, and rising economic nationalism.
- Trade regime heavily reliant on non-tariff measures; food self-sufficiency is a politically entrenched, securitised narrative that survives economic evidence to the contrary.
- MBG's unprecedented food procurement — poultry, eggs, vegetables, dairy, rice, and more, across 34 provinces — collides directly with this self-sufficiency agenda.

Effective rate of protection, 2015

(price + quantitative restrictions; Marks, 2017)

Commodity	ERP (%)
Rice	103.6
Sugar	87.6
Corn	27.9
Fruits	27.2
Vegetables	21.8
Livestock	16.3

Research Questions

Three Research Objectives

1. **Quantify** the economy-wide macroeconomic consequences of MBG — on prices, sectoral output, and household welfare — under three alternative policy scenarios: import protection, free trade, and productivity-led supply expansion.
2. **Distribute**: examine how these consequences are distributed across Indonesia's 34 provinces, identifying which regions bear disproportionate welfare costs or gains, and why.
3. **Prescribe**: establish the sectoral productivity improvements required to sustain MBG's nutritional objectives without relying on import restriction.

Gap addressed: the joint evaluation of a large-scale food demand program and its accompanying supply-side policy regime, at the sub-national level.

Model & Methodology

IndoTERM: A 34-Province CGE Model of Indonesia

Production structure

- ▶ Bottom-up, multi-regional CGE model; TERM family (Horridge 2012); previously applied to palm oil policy, Industry 4.0, capital city relocation.
- ▶ Nested CES: Leontief (intermediates + primary factor) → CES factors (land/labour/capital) → CES labour by occupation.
- ▶ Armington CES nest: domestic vs. imported varieties, at both industry and final-user level.

Demand & trade

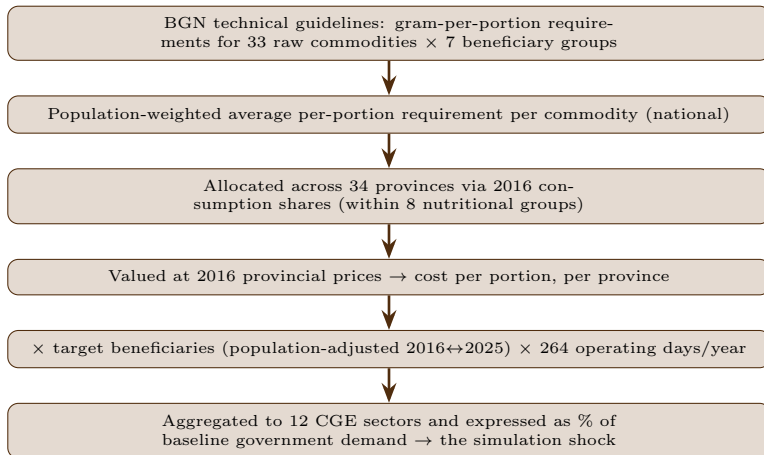
- ▶ Household demand: Klein–Rubin utility → linear expenditure system (subsistence + discretionary spending).
- ▶ Investment allocated by relative regional rates of return.
- ▶ Regional export demand: constant-elasticity schedules linking volumes to foreign-currency prices.

Database: 2016 National Input–Output Table + Interregional I–O Table + BPS regional production shares.

Model Closure: Short-Run

- ▶ Real investment and government demand exogenous (except the MBG shock itself) — isolates MBG's demand effect from fiscal feedback.
- ▶ **Labour:** real wages fixed nationally *and* regionally; national employment free to adjust (Keynesian-style closure, consistent with underemployment in food-producing regions).
- ▶ **Capital & land:** fixed within each region — no inter-regional mobility, no new investment within the simulation period.
- ▶ Small open economy, exogenous world prices; nominal exchange rate is the numeraire.
- ▶ National trade balance fixed as a share of GDP; regional household income linked to regional wage income.
- ▶ All technical change exogenous — none in Scenarios 1–2; calibrated TFP shocks imposed only in Scenario 3.

Calibrating the MBG Demand Shock



Three Policy Scenarios

S1: Protection

Imports of MBG-relevant commodities held at baseline; domestic tariff endogenized to clear markets. Tariff revenue not passed to the government budget.

S2: Free Trade

Imports free to adjust endogenously at world prices — the demand shock is partially absorbed through import expansion.

S3: Productivity

Sector-specific TFP improvements (grid search, uniform across provinces), sized so food imports remain negligible.

All three scenarios face the *same* MBG demand shock — what differs is how the supply side is allowed to respond.

National Results

National Macroeconomic Impacts

Indicator	S1 Protection	S2 Free Trade	S3 Productivity
<i>Price and consumption</i>			
Consumer Price Index	2.08	1.74	−0.15
Food CPI	5.74	4.53	−2.02
Real household consumption	−0.59	−0.25	1.79
Adjusted household consumption ^a	0.79	1.13	3.16
<i>Output and trade</i>			
Real GDP	−0.32	−0.14	1.51
Total imports	0.63	1.11	1.15
Total food imports	0.00	12.61	−0.05
<i>Welfare</i>			
Real GNE (C+I+G), national	0.47	0.67	1.86

^a Adjusted HH consumption adds the MBG in-kind transfer's real value at procurement cost — the appropriate measure vs. the no-MBG baseline. **Key takeaway:** protection taxes away much of the transfer's value (real consumption −0.59% despite adjusted welfare +0.79%); productivity is the only regime where the CPI falls outright and welfare more than triples protection's gain.

Sectoral Output Impacts (% deviation from baseline)

MBG-core food sectors

Sector	S1	S2	S3
Fruits	11.29	8.51	13.68
Poultry & Eggs	9.16	9.42	7.99
Dairy products	10.36	8.98	13.52
Abattoir	7.47	7.51	10.70
Processed soy	8.69	8.86	11.09
Vegetables	5.70	5.07	8.71
Rice milling	3.55	3.41	4.94
Cassava	0.92	1.10	2.51

Other food sectors

Sector	S1	S2	S3
Fish	0.08	0.23	3.50
Edible oils	-0.88	-0.75	2.62
Bread & biscuit	-0.52	-0.35	1.29
Pasta & noodle	-0.39	-0.22	1.46
Other food products	-0.90	-0.59	0.81

Non-food sectors

Sector	S1	S2	S3
Other agriculture	0.02	0.18	1.03
Other extractive	-0.29	-0.24	0.04
Other industry	-1.70	-1.41	0.30
Trade & services	-0.60	-0.35	1.05
Utilities	-0.47	-0.27	0.83

Key takeaway: MBG pulls resources into core food sectors across all scenarios, but under protection this comes at the cost of non-food output (other industry -1.70%, trade & services -0.60%); under productivity, every sector expands — there is no crowding-out.

National Consumer Price Impacts (% deviation from baseline)

MBG-core food sectors

Sector	S1	S2	S3
Fruits	23.01	14.81	-3.68
Vegetables	9.65	7.70	-1.71
Poultry & Eggs	6.14	5.96	-4.54
Dairy products	4.37	3.16	-0.90
Processed soy	4.09	3.67	-0.50
Abattoir	3.47	3.25	-3.62
Rice milling	3.60	3.20	-1.69
Cassava	3.05	2.95	-0.57

Other food sectors

Sector	S1	S2	S3
Fish	2.26	2.42	-2.27
Edible oils	0.55	0.48	-0.34
Bread & biscuit	1.64	1.42	-0.46
Pasta & noodle	1.84	1.61	-0.53
Other food products	1.83	1.59	0.11

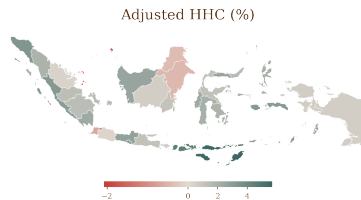
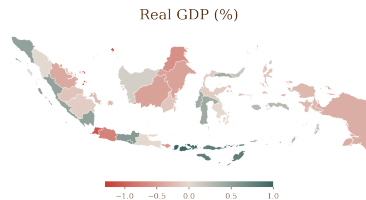
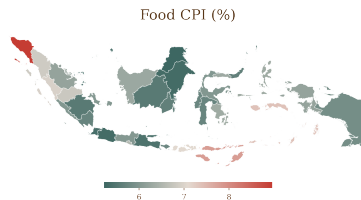
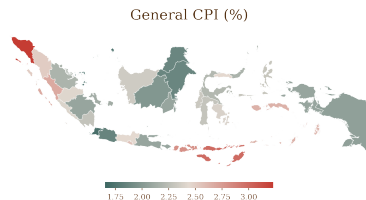
Non-food sectors

Sector	S1	S2	S3
Other agriculture	2.00	1.83	0.89
Other extractive	-0.12	-0.07	0.23
Other industry	0.63	0.56	0.17
Trade & services	1.44	1.27	0.31
Utilities	0.52	0.61	1.15

Key takeaway: Free trade eases prices for every MBG-core food relative to protection, though by varying amounts across commodities (e.g. fruits 23.01% → 14.81%; poultry & eggs 6.14% → 5.96%). Productivity is the only regime where every MBG-core food price falls below baseline.

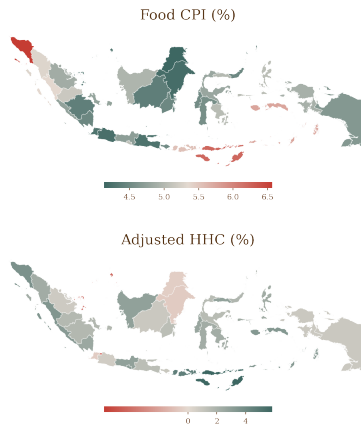
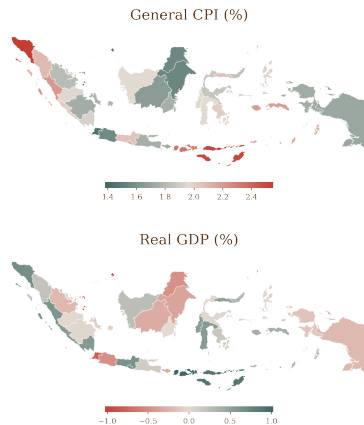
Provincial Results

Provincial Impacts: Protection



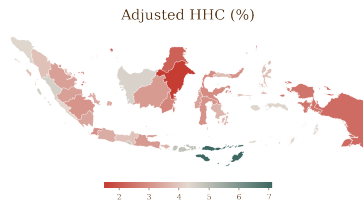
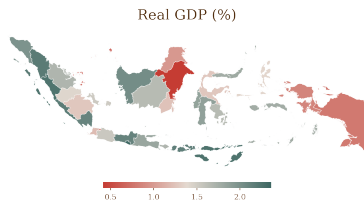
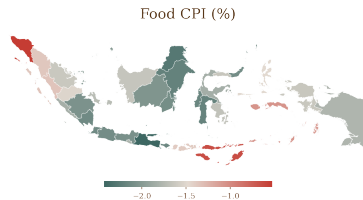
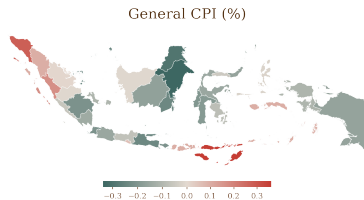
Highlights: NAD's large fruit demand shock drives the country's highest CPI (3.23%), yet still nets +3.37%. Only 5 provinces post net losses — DKI, KepRi, Banten, KalTim, KalUt — all combining modest local procurement with imported food inflation. NTT gains most (+5.68%). *Each map uses its own color scale.*

Provincial Impacts: Free Trade



Highlights: Same 5 provinces remain net losers as under protection, but losses are smaller throughout (e.g. DKI improves to -1.55%). Price increases ease nationwide relative to protection, and every province's welfare outcome improves. *Each map uses its own color scale.*

Provincial Impacts: Productivity



Highlights: All 34 provinces record positive adjusted welfare, ranging from +1.49% (KalTim) to +7.11% (NTT). General and food CPI turn negative almost everywhere — MBG now makes food *cheaper*, not more expensive. *Each map uses its own color scale.*

Who Loses Under Protection — and Why?

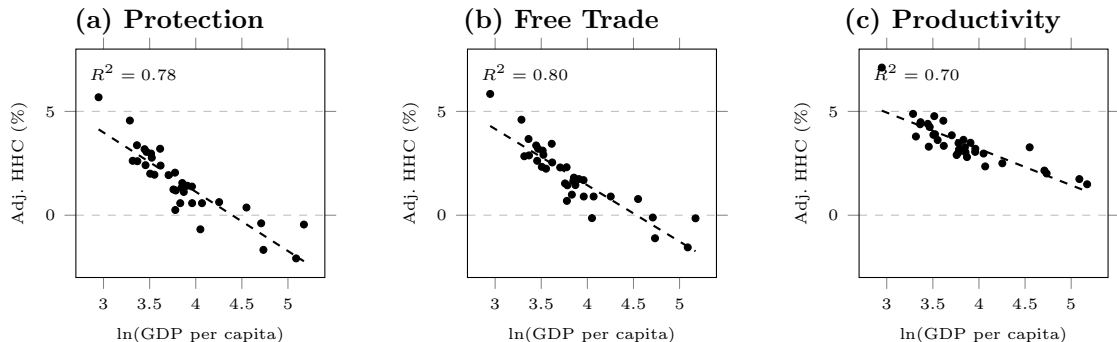
5 of 34 provinces record net losses

- ▶ DKI Jakarta: Adj. HHC -2.08% (CPI $+1.65\%$)
- ▶ Riau Islands (KepRi): -1.67% (CPI $+1.72\%$)
- ▶ Banten: -0.68% (CPI $+1.71\%$)
- ▶ East Kalimantan (KalTim): -0.45%
- ▶ North Kalimantan (KalUt): -0.39%

Common structural feature

- ▶ Modest local MBG procurement $\rightarrow$ small in-kind transfer.
- ▶ Yet exposed to food price inflation transmitted from producing provinces via inter-regional trade.
- ▶ Real GDP also contracts in all 5 (e.g. DKI -1.30% , KepRi -1.31%) — non-food, urban/service economies left out of MBG's shift toward food production.
- ▶ DKI: fruit demand shock only 3.98% , yet fruit prices rise 20.57% — driven by price transmission, not local demand.
- ▶ Contrast: NAD (highest CPI, 3.23%) still nets $+3.37\%$; NTT nets the highest gain of any province, $+5.68\%$, from its outsized dairy procurement share.

MBG's Progressive Geographic Incidence



Each panel plots adjusted household consumption (% dev. from baseline) against ln(provincial GDP per capita). Dashed line: OLS trend. Poorer provinces gain more, consistently across all three policy regimes.

Key takeaway: The gradient is nearly identical under protection and free trade ($R^2 = 0.78, 0.80$) — the progressive incidence is driven by MBG's procurement geography, not trade policy. Under productivity ($R^2 = 0.70$), even wealthier provinces gain substantially, narrowing the gap.

Conclusion

Key Findings

- ▶ MBG is a demand-side macroeconomic event, not merely a nutritional transfer: welfare depends critically on the accompanying trade regime.
- ▶ **Protection:** general CPI +2.08pp, food CPI +5.74pp; broad inflation taxes away much of the transfer's real value; adjusted welfare still positive (+0.79%) but 5 provinces are net losers.
- ▶ **Free Trade:** improves on protection (adjusted welfare +1.13%; smaller provincial losses) but requires food imports to rise 12.61% — at odds with the self-sufficiency agenda.
- ▶ **Productivity:** all 34 provinces gain (adjusted welfare +3.16%); food prices fall *below* baseline; food imports remain essentially unchanged (−0.05%).
- ▶ Progressive geographic incidence holds under all three regimes ($R^2 = 0.78, 0.80, 0.70$): poorer provinces consistently benefit more.

Why Is MBG Progressive? Decomposing the Mechanism

Indicator (Protection)	r vs. $\ln(\text{GDPpc})$	Reading
General CPI	-0.77	Poorer provinces face slightly higher inflation
Real GDP	-0.79	Poorer provinces gain output share as production reallocates toward food
Real HHC (no transfer)	-0.82	Combined price + output effect already favors poorer provinces
Adjusted HHC (+ transfer)	-0.89	The transfer amplifies the progressivity further

- (a) Poorer, less-connected provinces face *modestly* higher inflation — real, but a secondary effect.
- (b) Transfers are dramatically more valuable to poorer provinces: implied transfer value ranges from +0.4pp (DKI) to +4.3pp (NTT) of household consumption.
- (c) Richer provinces don't pay an *inflation* tax — they pay an **output-reallocation tax**: MBG shifts economic activity toward food-producing, mostly poorer provinces, contracting real GDP in richer, non-food, urban economies.

Net: (a)+(c) on the cost side are outweighed by (b) on the benefit side — progressivity is driven by *who produces the food and receives the transfer*, not by who pays more at the register.

Policy Implications

- ▶ The welfare impact of MBG depends as much on the *policy environment* as on the program design itself.
- ▶ Free trade is an immediately available, no-cost improvement — but politically incompatible with Indonesia's entrenched self-sufficiency narrative.
- ▶ Productivity delivers the best outcome on both welfare *and* self-sufficiency grounds, but feasibility varies by sector: required TFP gains range from 0.75% (bread & biscuit) to 10.00% (fruits) — most within Indonesia's historical productivity performance, fruits the exception.
- ▶ **Realistic near-term mix:** trade liberalisation for import-responsive sectors + targeted productivity investment (dairy, poultry, abattoir) where TFP requirements are attainable.
- ▶ The central question is not *whether* to pursue food self-sufficiency alongside MBG, but *through which instruments* — and at what pace.

Feeding a Nation

Thank you for listening!

Questions and comments welcome.